

STATUS: WARNING

SOP-002: Grid Failure & Genset Failover

Effective Date: 2026-02-01 Owner: Power Engineering Review Date: 2026-08-01

1. Purpose & Scope

This SOP applies to all DisCo/grid failure events within the Lagos metro region requiring automatic or manual genset failover to maintain tower uptime. Applicable to Lagos West, Ikeja, Lekki, Victoria Island, Ikoyi, Apapa, and Festac.

2. Step-by-Step Procedure

Step	Action	Owner	Time Target	Verification
1	Detect grid alarm	System	Immediate	Alarm in NOC dashboard
2	Auto-engage genset	System	<15 min	Controller status green + fuel pressure
3	If auto-genset inhibited -> manual start	field-team	<20 min	All towers back online
4	Notify IKEDC/EKEDC via NOC API	NOC	<30 min	Ticket acknowledged
5	Pre-position fuel for next 12h	Logistics	<60 min	Fuel delivery confirmed
6	If 2+ towers affected -> Tier-2 escalation + regulatory incident report	Management	<30 min	Escalation logged

3. Fuel Pre-Positioning Formula

Fuel Volume = duration_forecast x consumption_rate x 1.2 (buffer factor)

4. Common Failure Modes

Failure Mode	Mitigation
Maintenance flag inhibition	Ensure maintenance flags are cleared post-servicing
Fuel empty	Implement 12h reserve policy and monitor fuel levels via telemetry
Controller fault	Schedule quarterly controller diagnostics and firmware updates

5. Regulatory Requirement

File a regulatory incident report if 2 or more towers remain offline for more than 30 minutes due to grid failure.

6. Policy

All clusters with more than 2 towers on the same feeder must maintain a minimum 12-hour fuel reserve to ensure resilience against grid instability.

7. Version History

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Version	Date	Change Summary
V1	2025-03-01	Initial release
V2	2026-02-01	Updated failover timings, added fuel policy